Practice runs and evidence

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use repeated trials to prepare an evidence-based design explanation.

SUCCESS CRITERIA - I can...

1. Measure the actual course distance and run times.
2. Calculate average speed and describe variation.
3. Choose a final change supported by repeated evidence.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “trial” and “mean” to explain your model or result. Sentence starters are optional.

trial: one run or repetition of a test

mean: sum of the values divided by the number of values

evidence: observations or measurements that support an idea

01 Warm-up: bring back an earlier idea

A 50-tooth driver turns a 350-tooth gear.
Give ideal output/input torque and speed ratios and state the loss assumption.

Word help: torque = the turning effect of a force about an axis

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: mean time = sum of trial times/number of trials
average speed = distance/time.
Measure course length; keep release and timing consistent.
Retain every trial and required build evidence.

Worked reasoning: Supplied 3 m runs take 2.0, 2.4, 2.1 s. Report all trials; their mean time is 2.17 s. “Fastest run 2.0 s” answers a different question from typical performance.

Your turn: Make a trial table and mark both mean and fastest time.

Sentence starter: I notice ... . This supports ... because ... .

Trial	Distance (m)	Time (s)	Speed (m/s)
1	____	____	____
2	____	____	____
3	____	____	____

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Complete a final labeled sketch or attach a photo with labels.
Identify a feature that supports straight motion and one that supports reliable electrical operation.

Sentence starter: The … in my model represents … . Its label shows … .



Marzano check - circle: 1 2 3

04 Apply the physics

A team’s supplied 4.0 m run times are 5.0, 5.2 and 4.8 s. Fill the table and calculate mean time, the full time interval, its width and speed calculated from mean time. Retain every trial.

Sentence starter: I found … [unit]. This value means … in this situation.

Trial	Distance (m)	Time (s)	Speed (m/s)
1	____	____	____
2	____	____	____
3	____	____	____

Marzano check - circle: 1 2 3

05 Explain your reasoning

One run stops early after a loose connection. Should you erase it?
Explain how to record it fairly and decide whether it belongs in a speed comparison.

Sentence starter: Under the assumption …, the evidence … supports … because … .

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

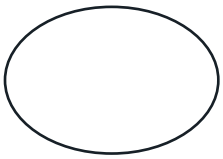
06 Review: connect an earlier idea

The same flux change through the same coil occurs in 1/4 of the time.
Compare induced-emf magnitude and name a condition for that comparison.
Label the field or flux and identify what changes.

Word help: flux = a measure of magnetic field passing through a chosen area

Sentence starter: Both ... . They differ in ... because ... .

Label field direction, loop area and what changes.



Turns: ____ Time interval: ____ emf: ____

Marzano check - circle: 1 2 3

07 Exit check

Write a three-sentence claim-evidence-reasoning explanation for a design choice. Distinguish a controlled comparison from a plausible mechanism only.

Sentence starter: My design shows ... . It meets the requirement by ... .

Choice / claim	Actual evidence	Physics explanation
____	____	____
____	____	____

Marzano check - circle: 1 2 3

08 STRETCH: test the model

For 3 m runs of 2.0 and 3.0 s, compare mean of the two speeds with total distance divided by total time.
Explain why the results differ.

Sentence starter: Both ... . They differ in ... because ... .

Trial	Distance (m)	Time (s)	Speed (m/s)
1	____	____	____
2	____	____	____
3	____	____	____

Marzano check - circle: 1 2 3